Aerodynamic Structural Analysis of An Aircraft Wing

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Abstract

Aerodynamic problems in general are often difficult to solve by analytical analysis. Experimental or numerical simulation can be used to analyze these computational models. However, due to the large expenses required in the experimental method, the numerical method is more preferred. This project presents the modelling and simulating processes of computational fluid dynamic (CFD) problem on a aircraft wing model, using typical section of NACA 2412 airfoil. ANSYS Fluent is used to analyze the pressure and velocity distribution on the surface of wing. In this structural analysis aircraft wing is fixed as a cantilever beam and pressure load is applied. The pressure force is imported from the results from the CFD analysis. The pressure that flow over the wing in the CFD analysis is applied as the pressure load on the top and bottom surface of the wing. This is done using software ANSYS FLUENT 15.0. The lift and drag forces are also determined by ANSYS Structural. Also stress distribution on the wing is studied for different materials with reference of journal papers and ANSYS is chosen for modelling the airfoil and analysis. The structural analysis is also compared with the different aircraft wing materials made of CFRP T300/976 and aluminium 7075 T6 for better strength. ANSYS FEA tool is used for the purpose of study. A further ANSYS FLUENT is needed for computational fluid dynamic analysis of aircraft wing model. Modelling, meshing and analysis is done ANSYS FEA tool.

2.Introduction

The authentic foundation of an auxiliary system of an airplane illuminates us about the past review regarding aeronautics all in all. The improvement in the substances and strategies consolidated to amass an air ship has influenced its change of the basic wooden defensive layer system to the smooth and rich streamlined flight motors of today. The primary discoveries that "raising" could be delivered by moving the air over a bended surface to get the development of fixed and pivoting planes moving. George Cayley made a powerful streamlined profile and later in that century effective kept an eye on lightweight flyers in the nineteenth century. He set up the standards of flight, including lifting, weight, push and obstruction. In this task, we will demonstrate a geometry

of a surface streamlined wing NACA 2412 of 1.6 m long and submit it to a CFD recreation to recognize the appropriation of weight above and underneath the wing by blowing air at a flat out speed of 50 m/s. The auxiliary investigation is performed on it to distinguish the most extreme pressure initiated in the structure of the wing dependent on the various properties of the materials and the structure of the wing is liable to various approaches to consider its adjustment in height, dissemination of weight and actuated exertion. Prior to examining the technique, the improvement of the limited component model and the contextual analysis, let us know quickly the fundamental structures, the kinds of pressures and the setup of the wing structure in the segments underneath.

3.Literature Review

The procedures of demonstrating and recreation of the issue of computational liquid elements (CFD) in a plane wing model, utilizing a common segment as a thought of NACA 2412 streamlined profile, are all around clarified in the accompanying record. This wing model could be picked later on trial plan. ANSYS Fluent is utilized to examine the weight and speed dissemination on the wing surface. Lifting and hauling powers are additionally controlled by ANSYS Structural. Also, the coefficients of the lifting and hauling powers can be determined through the information got when the relative speed contribution between the wind stream and the bearing surface changes from 0 to 50 m/s. The numerical outcomes indicated are perfect with those of the hypothesis, recommending a dependable choice to foresee the streamlined attributes of the wing model tried in the assembling of unmanned aeronautical vehicles (UAVs). Streamlined investigation of the wing of the Nguyen Minh Triet * air ship, Nguyen Ngoc Viet, Pham Manh Thang Falcuty of Mechanics and Automation, UNV University of Engineering and Technology, 144 Xuan Thuy, CauGiay, Hanoi, Vietnam, this report presents the idea and methods of CFD simulation and structural analysis for carrying out the project.

Warning Committee of the National Library of Aeronautical Technology kafb, nm mms01 specialized note where. I276 LOW SPEED TESTS OF FIVE 66-SSRIES AIRCRAFT WITH MEDIUM LINES DESIGNED by Albert E. von Doenhoff, Louis and James M. O'Connor. This book helps in designing the airfoil cross section and chamber length

CFD Analysis of UAV Flying Wing Vasile PRISACARIU Corresponding creator "Henri Coandă "Foundation of the Air Force of Braşov, Romania, Mihai Viteazul, nr. 160, Brasov 500187, helps in creating wing box CFD analysis and getting pressure distribution around wing and walls of the air box with absolute velocity of 50m/s.

S. Patel, Saumil B. Patel, Utsav B. Patel, Prof. Ankit P. Ahuja UVPCE, University of Ganpat. We gained drag and lift powers utilizing CFD in this examination, which can likewise be resolved utilizing wind passage test tests.

Universal Journal of Engineering and Technology Research (IRJET) e-ISSN: 2395-0056 Volume: 04 Edition: 06 DESIGN AND ANALYSIS OF FINAL AIRCRAFT WINGS USING TAPE AND SPARS Sivarama Prasad Peruru, Suman BabuAbbisetti. This report can be utilized as a source of perspective to show the structure of the wing with the expansion of ribs and spars in future work.

4. Finite Element Model Development

As the quickest methods for transportation accessible, the plane has turned out to be progressively mainstream lately. This notoriety has brought about the advancement of quicker and more secure planes by numerous analysts. Numerous instruments dependent on CFD investigation and numerical techniques have been made and can be helpful for streamlined examinations in a flying machine.

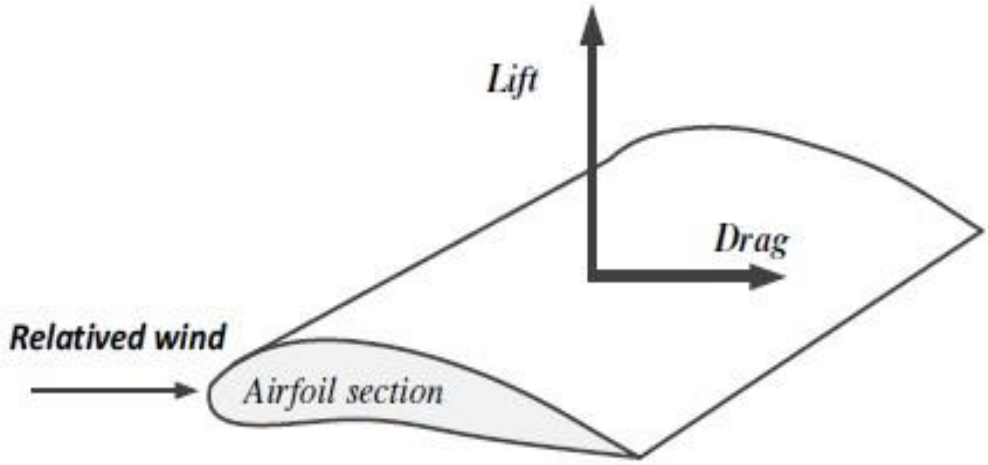

Fig 2.1 Geometry of an aircraft wing.

Streamlined features is an all-encompassing field of mechanics, which concentrates the powers and minutes important for feasible air development. Streamlined powers following up on the flying article are called rise the typical way of flight and drag power or impetus toward flight (see Fig. 2.1). These powers rely upon the speed of the stream well in front of the article, which is called relative breeze. The assurance of the collaboration between the air and a solid body that moves in it, similar to the wing of a plane, is identified with a wide range of streamlined issues. Streamlined scientists have been keen on the ideal types of the streamlined surface, to give the most noteworthy height and least protection from the wing during departures and during the flight.

A streamlined profile is characterized as the condition of a wing as found in the cross-sectional region. The fundamental plan of the streamlined surface is appeared in Fig. 2.1. The principal methodical investigation of the states of the streamlined surface and their exhibition was worked by the National Aeronautics Advisory Committee, NACA (an antecedent to NASA). The rope, the rope and the thickness are the most significant attributes of streamlined geometry. The exhibition attributes that inventoried streamlined profiles for the most part give incorporate height, opposition, weight circulation and timing of the streamlined focus. The breeze passage tests assessed the estimations of these qualities and are likewise decided through scientific hypothesis investigation or utilizing computational models with CFD reenactment apparatuses. In this record, the ANSYS

programming plays out the CFD investigation on the flying machine wing model utilizing the NACA 2412 profile. The target of this investigation is to test the produced wing model, which could be utilized to plan future air ship.

4.1 Problem definition.

In this task we are concentrating the conduct of the flying machine wing under the weight load for the investigation of static auxiliary opposition. The streamlined geometry NACA 2412 is chosen to show the wing. The 1.6 m wing length is utilized for investigation in this venture. The issues and their advancement of limited components associated with our undertaking are talked about underneath.

1. Streamlined issues when all is said in done are frequently hard to tackle by expository examination. Test or numerical recreation can be utilized to examine these computational models. Be that as it may, because of the enormous costs required in the exploratory strategy, the numerical technique is increasingly liked.

2. The principle issue confronted today in the airplane business is to create lightweight and high quality air ship structures. This undertaking includes the examination of the flying machine wing made out of 7075-T6 aluminum amalgam and CFRP T300/976 composite material for better static basic quality of the plane wing flexed.

3. This article exhibits the demonstrating and reenactment procedures of the computational liquid elements (CFD) issue in a plane wing model, utilizing the regular segment as the NACA 2412 streamlined profile. This wing model could be picked later on exploratory plan .

4. ANSYS Fluent is utilized to investigate the weight and speed conveyance on the wing surface.

5. The CFD weight information is utilized to perform static auxiliary dissects on the wing of a flying machine that has a NACA 2412 profile area.

6. In this task, two 7075-T6 aluminum combination and CFRP T300/976 unidirectional materials are utilized for the investigation.

7. Instrument utilized: Ansys programming - Ansys Workbench, Ansys Fluent and Ansys Structural.

4.2 Problem Statement

The task includes displaying the wing of the air ship with a cross area of streamlined surface, for example, NACA 2412 and a length of 1.6 m. The 5 m wide and 3 m high air box is utilized for CFD investigation as appeared in Figure 2.2. The air at an outright speed of 50 m/s enters the passage of the air box and, because of the plan of the streamlined sheet, high weight and low weight are made on the wing. This conveyance of weight on the wing is plotted and the outcome is moved to the Ansys structure for the basic examination of the wing where a part of the bargain that goes about as a cantilever pillar is fixed.

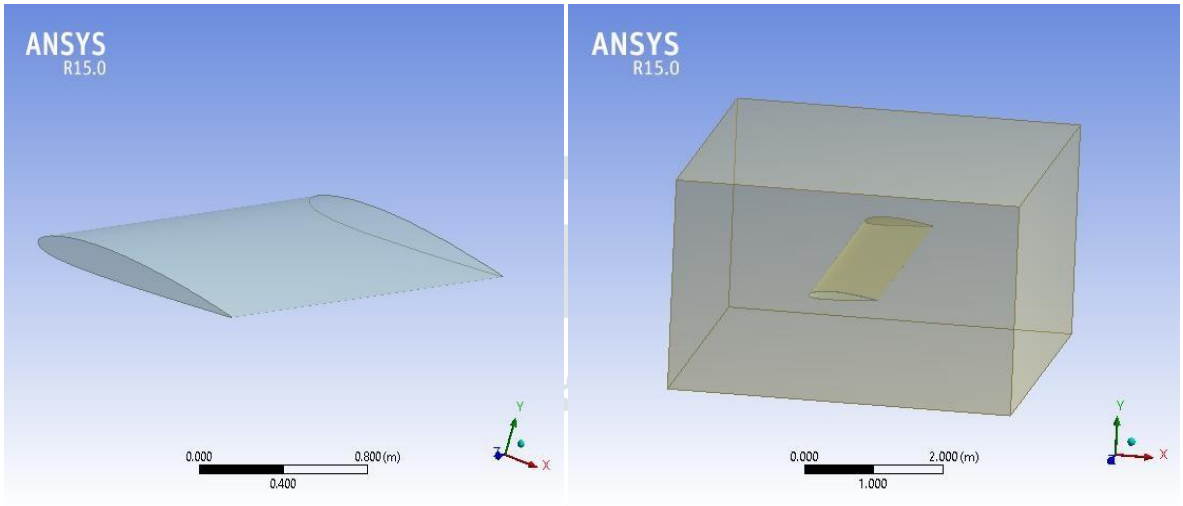

Fig 2.2 wing model and airbox modeled using ansys in the project.

4.3 Material choice

Metallurgy has assumed a significant job in the development of flight. As of not long ago, some new materials, for example, titanium amalgam or compound, have been actualized in flying machine development. In any case, these superalloys are still very costly for the flying machine developer. With its focal points in weight and cost proportion, aluminum composite is still broadly utilized. 7075 T6 aluminum compound is likewise an appealing material, so it is utilized in the plan of this investigation. The correlation between the unidirectional T300/976 CFRP and the 7075 T6 aluminum combination is likewise performed to break down the most extreme anxieties actuated in the wing for two distinct materials.

4.4 Engineering Data

Designing information is utilized to choose a material properties of material utilized for investigation. load the alluminium composite 7075 T6 and CFRP T300/976 for two distinctive examination. For entering the CFRP material properties go to make new material select straight enter youngs modulus 230

Gpa and toxin's proportion 0.3 and mass thickness of 1600 kg for each cubic meter. Material properties utilized for examination is appeared in table 3.1.

	Aluminium alloy 7075 T6		CFRP T300/976	
Material property	value	unit	value	unit
Mass density	2810	kg/m3	1600	kg/m3
Young's Modulus	71.7	Gpa	230	Gpa
Poison's ratio	0.3		0.318	

Table 3.1 material properties of materials used in the project.

5.CASE STUDY

In this venture we will show a NACA 2412 airfoil wing geometry of length 1.6m and exposing it to CFD reenactment to distinguish the weight dissemination above and underneath the wing by blowing air at total speed of 50m/s by keeping the wing inside an airbox, which is displayed. auxiliary examination is done on the equivalent to recognize the maximum pressure initiated in the wing structure for various materials and the wing structure is exposed to various approaches for considering its adjustment in lift, weight dissemination and stress actuated. Framework coupling is utilized to associate the CFD results to auxiliary examination where CFD weight burden is taken as the limit conditions and connected on the outside of wing utilizing liquid strong interface condition. For CFD investigation explicit limit conditions must be set where speed is supreme, stream is laminar and weight based examination. The wing is considered as the cantilever shaft where one end is fixed to the fuselage as a limit condition for the basic examination. The wing structure is exposed to various point of assaults for examining weight circulation in CFD recreation. The lift created, drag, weight conveyance, stress appropriation and most extreme pressure incited for wing made of two unique materials which are CFRP and Aluminium compound for various edge of attacks are examined beneath. The outcomes are quickly talked about in this segment.

5.1 Graphical Post Processing and Result Discussion

Most extreme weight and least weight initiated, lift, drag and greatest pressure actuated for various materials utilized in this venture are appeared in forbidden section 4.1

Angle of attack	Lift (N)	Drag (N)	Maximum pressure (pa)	Von misses stress (pa)	Total deformation (m)	Strain (m/m)
0^0	175.2	4.3N	1.44e3	4.00e6	0.0019063	0.00011
1^0	201.3	7.594	1.42e3	5.15e6	0.0021767	0.00012
4^0	354.89	10.10	1.39e3	9.61e6	0.0045795	0.00013534
8^0	533.03	25.06	1.54e3	1.44e7	0.006896	0.00020289

12^0	579.036	54.2	1.58e3	1.58e7	0.00759	0.00022365
16^0	693.8	141.94	1.68e3	1.73e7	0.0082869	0.00024378
20^0	-10.1375	66.4	1.49e3	1.52e7	0.0026537	0.00000759

Table 4.1 Results of aluminium alloy wing under different angle of attack.

Angle of attack	Lift (N)	Drag (N)	Maximum pressure (pa)	Von misses stress (pa)	Total deformation (m)	Strain (m/m)
0^0	175.2	4.3N	1.44e3	3.92e6	0.000532	1.563e-5
1^0	201.3	7.594	1.42e3	4.225e6	0.0006469	1.90e-5
4^0	354.89	10.10	1.39e3	9.52e6	0.0012048	3.528e-5
8^0	533.03	25.06	1.54e3	1.43e7	0.0018155	5.298e-5
12^0	579.036	54.2	1.58e3	1.57e7	0.001997	5.85e-5
16^0	693.8	141.94	1.68e3	1.723e7	0.0021852	6.3838e-5
20^0	-10.1375	66.4	1.49e3	1.49e7	0.0002432	6.9982e-5

Table 4.2 Results of CFRP wing under different angle of attack.

From the tabular column it is observed that as the angle of attack increases max pressure induced increases and lift also increases. At 20^0 angle the lift is negative and drag force is more. As the angle of attack increase drag force on the wing also increases so it is important identify the angle of attack for which optimum lift is produced. It is evident from the results that for better lift the wing should be designed between 0 degree to 16 degree based on the required lift for aircraft. The following figure shows the pressure distribution for wing under 1°, 4°, 8° and 16° angle of attack.

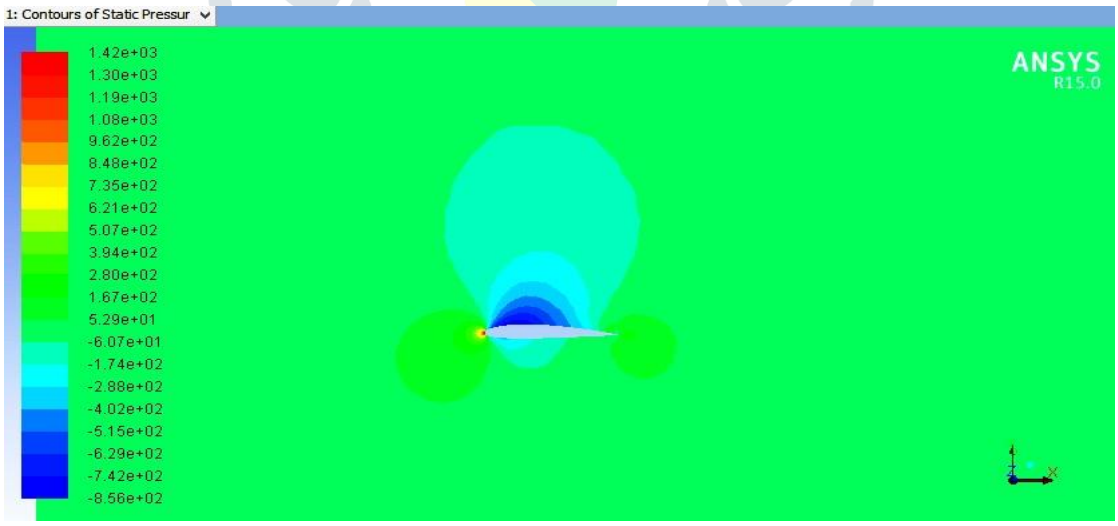

Fig 4.1 pressure distribution for 1° angle of attack

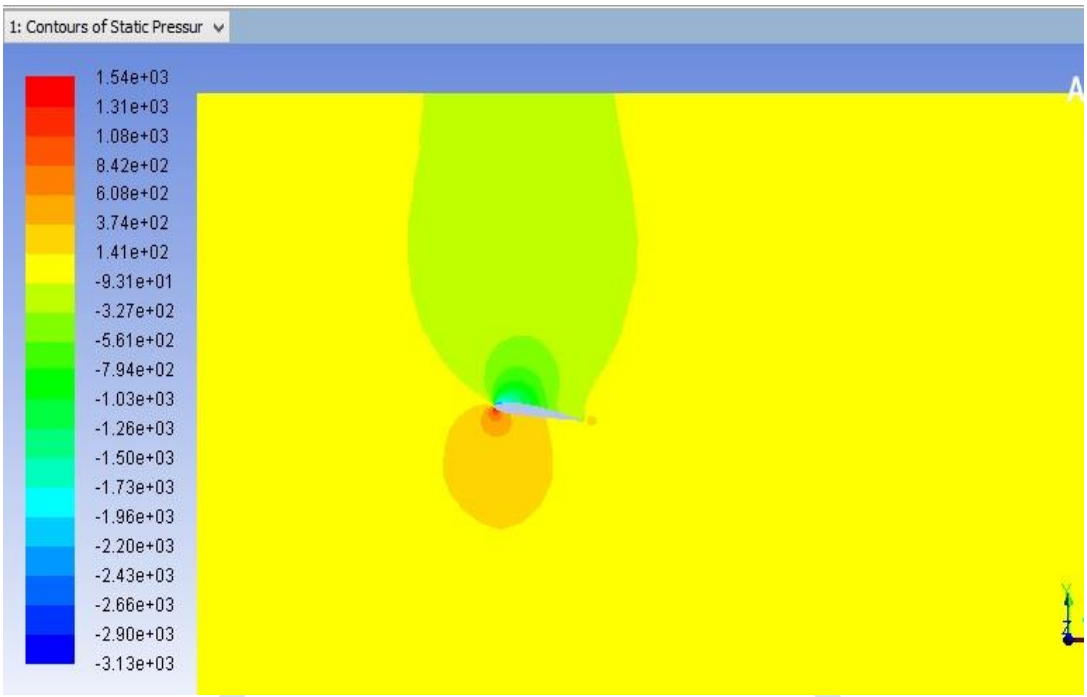

Fig 4.2 pressure distribution for 4° angle of attack

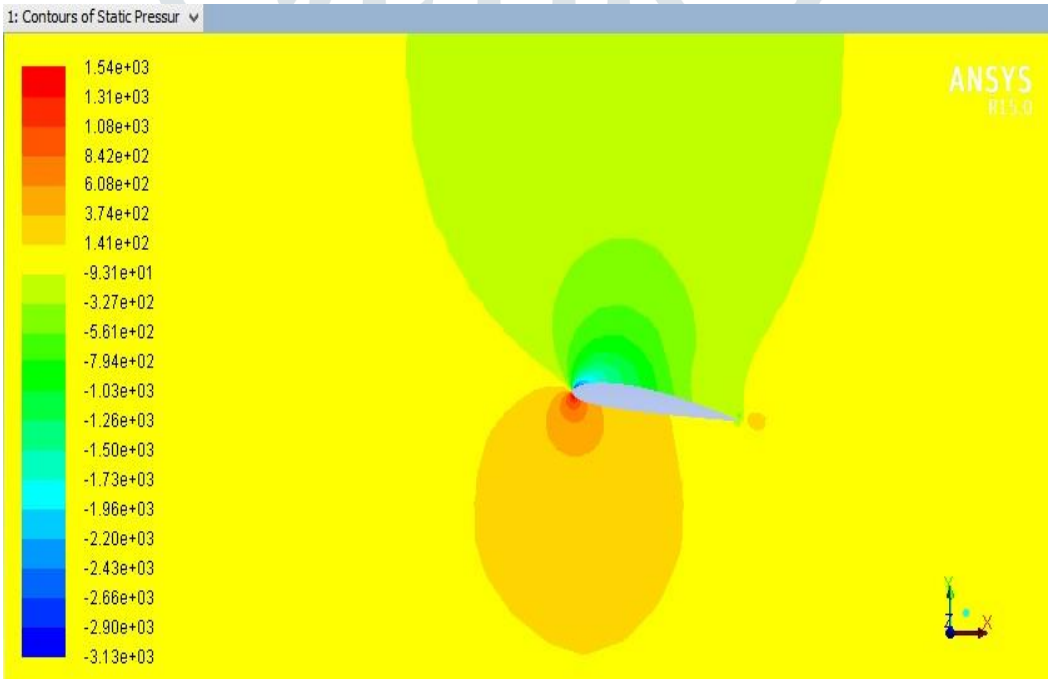

Fig 4.3 pressure distribution for 8° angle of attack

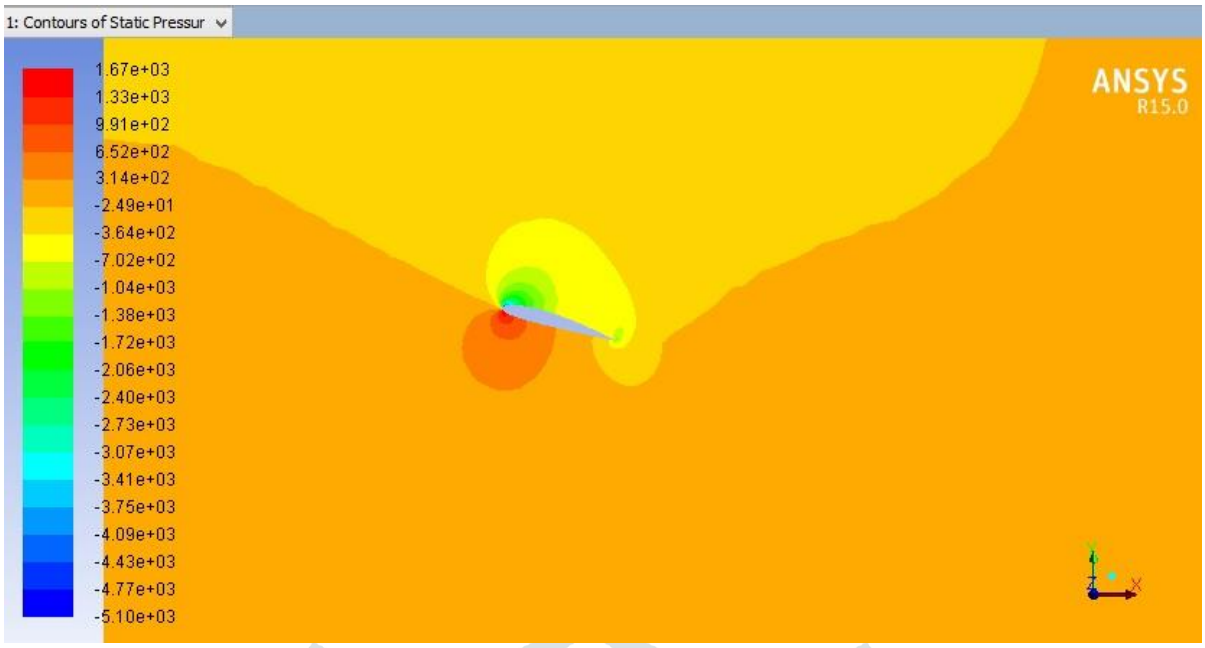

Fig 4.4 pressure distribution for 16° angle of attack

The maximum pressure induced for the different angle of attacks, lift and drag are tabulated in the tabular column 4.1 and 4.2. As the angle of attack increased the increase in pressure, lift and drag is noticed. Equivalent von mises stress and deformation induced in aluminium wing

CFRP wing for different angle of attack is shown below.

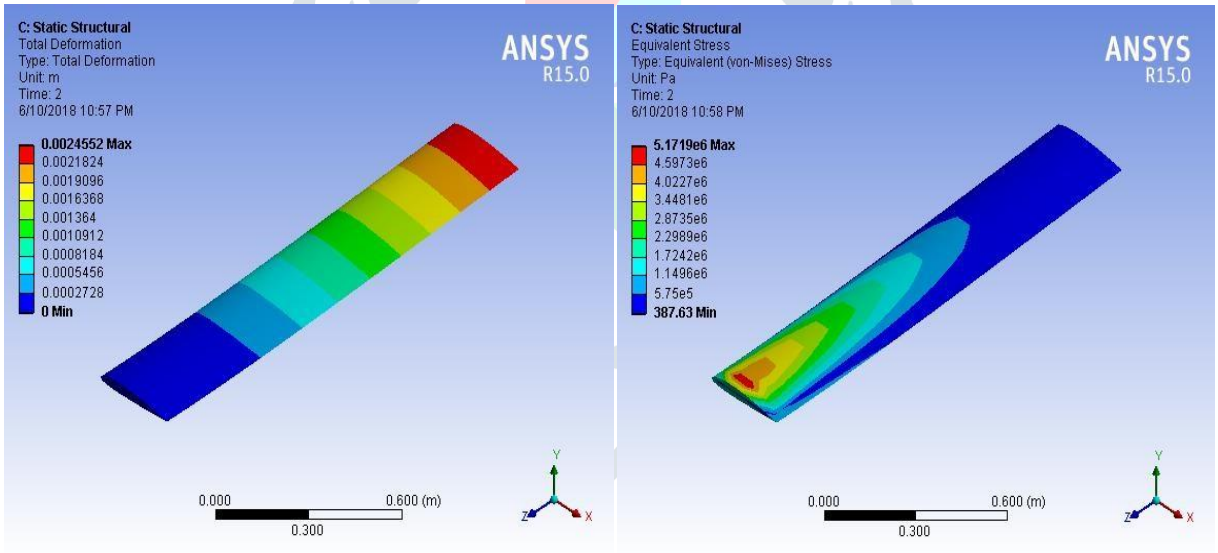

Fig 4.5 Deformation and stress induced in aluminium alloy wing for 1° angle of attack

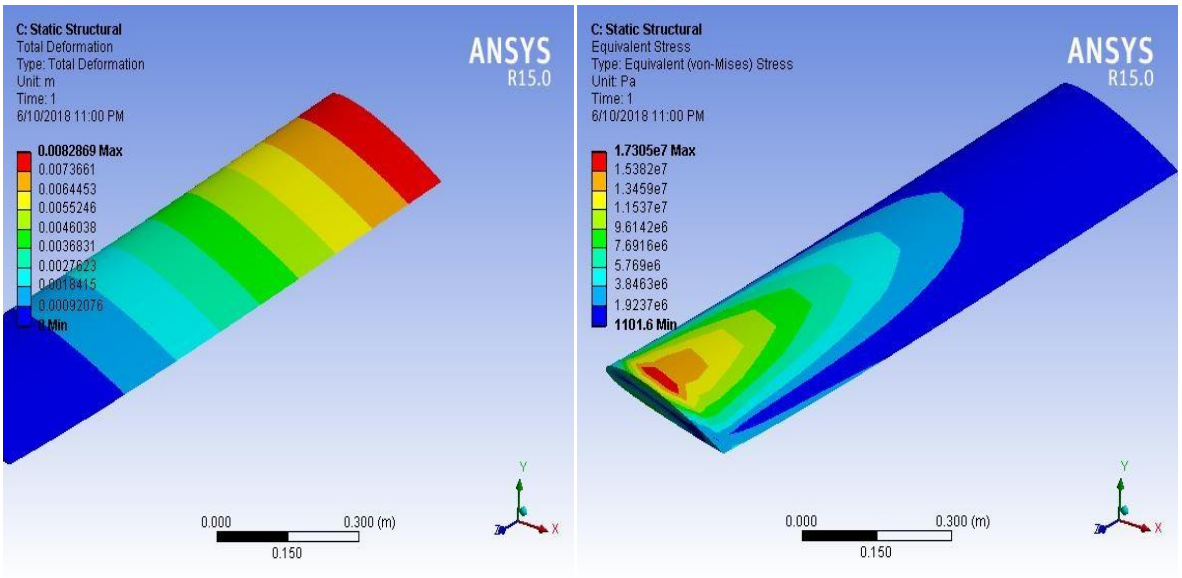

Fig 4.6 Deformation and stress induced in aluminium alloy wing for 16° angle of attack

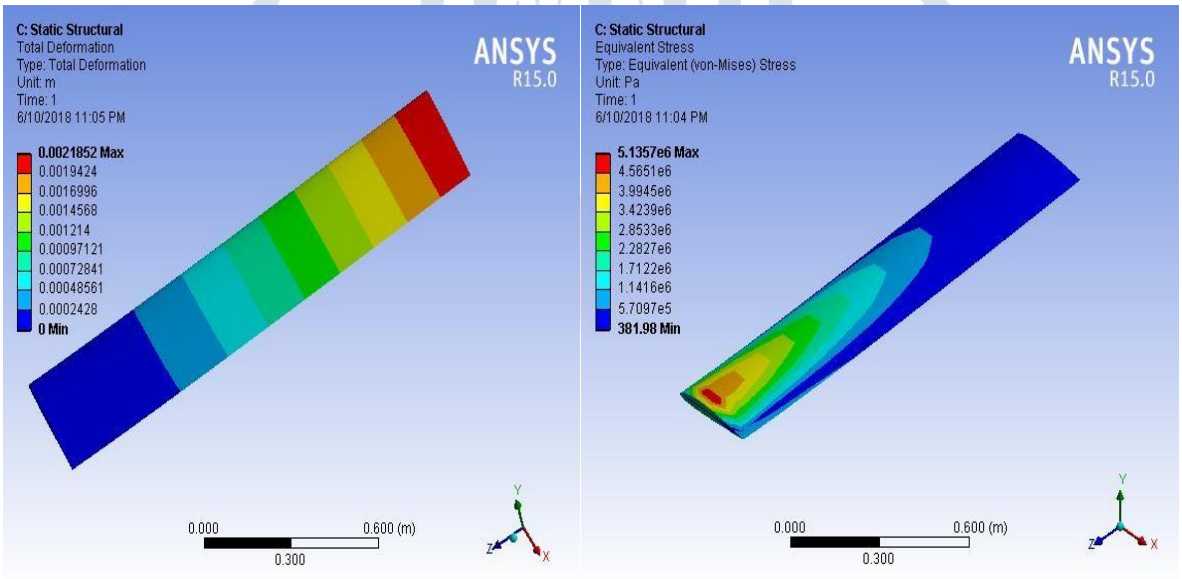

Fig 4.7 Deformation and stress induced in CFRP wing for 1° angle of attack

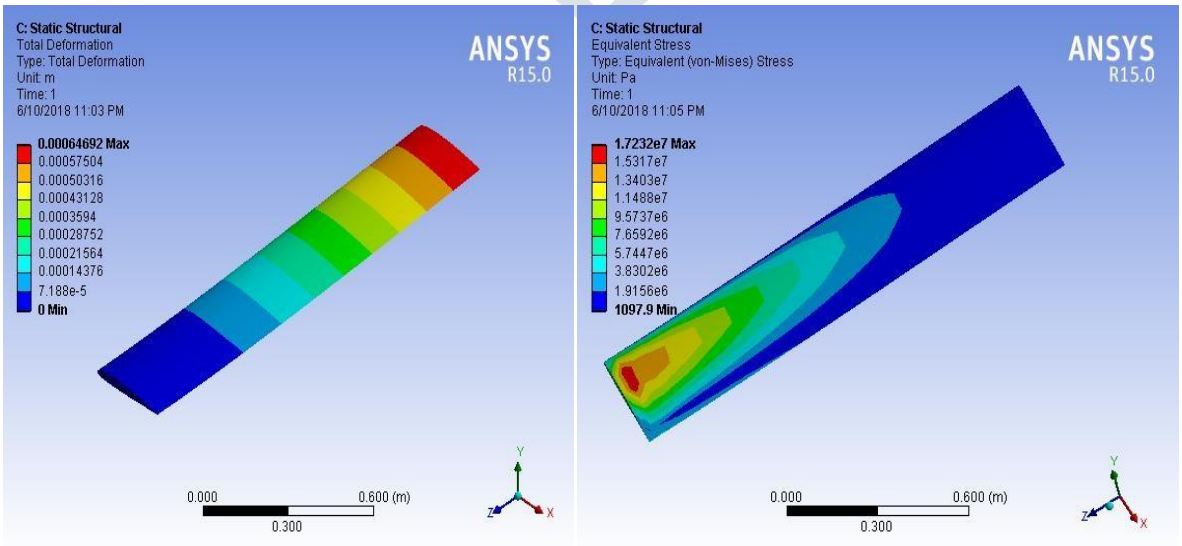

Fig 4.8 Deformation and stress induced in CFRP wing for 16° angle of attack Result obtained from the analysis shows that the stress induced in the CFRP wing material is less than the Aluminum

alloy wing material. It concludes that the CFRP can be used to make wing structure than alluminium since stress induced in CFRP is less. Deformation due to pressure load on the wing is also less in CFRP compared to alluminium. So CFRP has more stiffness and can withstand deformation due to pressure load. These results can be used in conducting a static failure analysis and as a reference for modeling and designing a wing structure. This project involves modeling a wing of 1.6 m length and CFD analysis and structural analysis on same, so this results are best suited for low weight aircraft designing.

6.Conclusion and Future Work

6.1 Conclusions

- The maximum lift produced at 16 degree angle of attack is 693.8 N but drag force is 141.94 which is more considered to 12 degree angle of attack. Where lift produced at 12 degree angle of attack is 579 N and drag is 54.2 N which is very less. So the thrust required for lifting wing with 12 degree angle of attack is less as drag force is less compared to wing with 16 degree angle of attack where drag force is more.
- Pressure distribution, lifts produced and drag results obtained from analysis helps in choosing better angle of attack for wing aircraft structure based on the requirements. These results can be used in design and modeling of low aircraft structures like unmanned vehicles.
- The loads of CFD simulation is applied on the wing in the structural analysis using system coupling. The stress induced in the CFRP for different angle of attack is less compared to aluminium 7075 wing. Stress induced in CFRP wing at 12 degree is 1.57e7 and aluminum wing is 1.58e7. Where deformation induced in CFRP under 12 degree is 0.001997m and aluminium wing is 0.00759m.
- It is observed from the results that stress induced and deformation is less in the CFRP wing than the aluminium wing. So the stiffness of the CFRP is more as well as strength is also more compared to aluminum wing. These results can be used for analyzing the better material and failure stress analysis.

6.2 Future Work

- The wing can be modeled with spar and ribs and can be subjected to structural analysis keeping this project results as reference. It helps in changing the wing structure design and wing can be modeled with beams and can be subjected to structural analysis.
- The results from structural analysis can be used to conduct failure stress analysis in CFRP and aluminum wing structure. Stress results from the analysis can be taken as reference for determining safe loading conditions and failure analysis. The procedure can be used to identify the better material than the materials used in this project.

7.REFERENCES

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